

Public whole-genome demonstration

HG004 · Genome in a Bottle

Sanitized public demonstration generated from an end-to-end GenomeLens analysis of publicly released Genome in a Bottle data.

This public report contains a selected allowlist of results from the analysis. It is not the production GenomeLens report and does not expose the private analysis engine.

DATASET PROVENANCE

SOURCE Genome in a Bottle	REFERENCE BUILD GRCh38	BENCHMARK RELEASE v4.2.1	BENCHMARK SCOPE chr1-22
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EXTERNAL BENCHMARK

CONCORDANT 170/170	EXPLICIT TRUTH CALLS 170	MISMATCHES 0	NORMALIZATION FAILURES 0
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Concordance is measured on positions where the truth set states a genotype, inside the defined confident regions. It is not whole-genome accuracy, not clinical validation, and not an endorsement by NIST or GIAB.

SELECTED GENOMIC RESULTS

FINDING	CATEGORY	DISPOSITION	EVIDENCE BASIS
Below-median longevity composite (improvement headroom)	longevity	conditional	published association evidence
CYP2C9 Intermediate Metabolizer (IM) (warfarin)	pharmacogenomics	conditional	published pharmacogenomic guideline evidence
Elevated coronary-artery-disease polygenic risk	polygenic risk	conditional	published association evidence

A representative selection, one per category. The full sanitized analysis contains more results than are shown here.

HEALTH-ECONOMIC RESULTS

REFERENCE-CASE NMB \$138.56	CANONICAL EXPECTED NMB \$765.00	INCREMENTAL COST \$588.49	INCREMENTAL QALYS 0.0073
ICER 80,943	PROBABILITY COST- EFFECTIVE 49.4%	NOTE Reference-case and canonical pathway results answer different questions and are never summed.	

Evidence basis. Genomic findings rest on published association evidence; published pharmacogenomic guideline evidence. The economic figures are modeled estimates built on published evidence and stated assumptions, not realized savings.

For nontechnical readers. These results are a research demonstration of what GenomeLens can produce from real whole-genome data. The benchmark measures agreement only at the specific GIAB locations tested and is not a claim of whole-genome or clinical accuracy. The health-economic figures are modeled estimates based on published evidence and assumptions; they are not a diagnosis, medical advice, or a statement that a genome is literally "worth" a particular dollar amount.

SCOPE AND LIMITATIONS

- Public demonstration output; not clinical validation.
- Benchmark scope is limited to the stated GIAB panel and chr1-22.
- Modeled economics are not realized savings.
- Representative findings only; the full sanitized report is the complete output.

Methods and validation: [VALIDATION.md](#) · [ECONOMICS.md](#) · [LIMITATIONS.md](#)

Research software. Not a diagnostic device. Not medical advice.